

Public Key Cryptography

Key Generation

1. Algorithm Selection: Choose a cryptographic system (e.g., RSA, ECC).

RSA

?

ECC

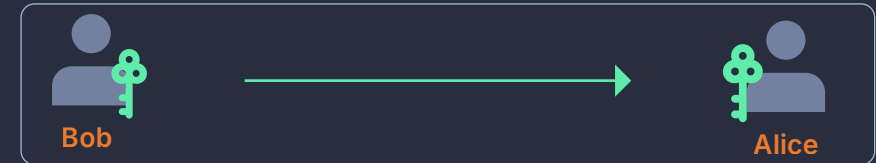
2. Generate Keys

Public Key: (e, n)
(shared openly)

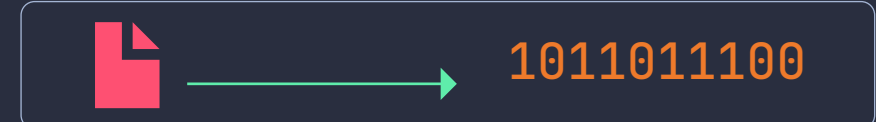
Private Key: (d, n)
(kept secret)

Encryption

1. Get Public Key: Alice retrieves Bob's public key (e, n).



2. Convert Message: Convert plaintext (M) to an integer m (e.g., using ASCII).



3. Encrypt: Compute ciphertext $c = m^e \bmod n$.



4. Send: Transmit c over an insecure channel.



Decryption

Receive Ciphertext: Bob receives c.



Decrypt: Use private key (d, n) to compute $m = c^d \bmod n$.



Convert Back: Translate integer m to plaintext (M).



Digital Signatures

The are used for Proving Authenticity

1. Hash the Message: Compute a hash digest $H(M)$ (e.g., SHA-256).



2. Sign: Encrypt the hash with Alice's private key: $S = H(M)^d \bmod n$.



3. Send: Transmit $M + S$ to Bob.



4. Verify: Bob decrypts S with Alice's public key and checks if it matches $H(M)$.

